



DELHI METRO RAIL CORPORATION LIMITED

(A JOINT VENTURE OF GOVT. OF INDIA AND GOVT. OF NCT OF DELHI)

DESIGN, MANUFACTURE, SUPPLY, INSTALLATION, TESTING AND COMMISSIONING OF ENVIRONMENT CONTROL SYSTEM (ECS) AND BUILDING MANAGEMENT SYSTEM (BMS) FOR 3 U/G STATIONS (HAUZ KHAS, MALVIYA NAGAR AND SAKET) AND ASSOCIATED TUNNEL SECTIONS ON CENTRAL SECRETARIAT TO QUTAB MINAR CORRIDOR AND 4 U/G STATIONS (CENTRAL SECRETARIAT, KHAN MARKET, JLN STADIUM AND JUNG PURA) AND ASSOCIATED TUNNEL SECTIONS ON CENTRAL SECRETARIAT TO BADARPUR CORRIDOR OF PHASE-II OF DELHI MRTS PROJECT

CONTRACT AGREEMENT BE - 12 LOT 3 & BE - 14 LOT 3

VOLUME 2

DELHI METRO RAIL CORPORATION LTD.
BCC Place, Bhishma Pitamah Marg,
Kirti Vihar, New Delhi - 110 003

M/s BLUE STAR LIMITED
E-44/12 Okhla Industrial Area,
Phase-II, New Delhi-110 020



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DELHI METRO RAIL CORPORATION LIMITED

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SCOPE OF WORK
SPECIFICATIONS



DELHI METRO RAIL CORPORATION LTD.
NBCC Place, Bhishma Pitamah Marg,
Pragati Vihar, New Delhi – 110 003



Sanjay
Sanjay



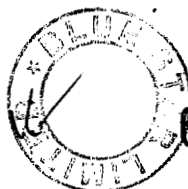
DELHI METRO RAIL CORPORATION LTD.
MASS RAPID TRANSIT SYSTEM -PHASE-II

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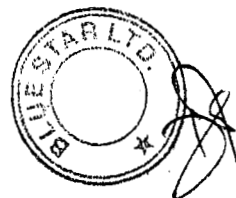
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VOLUME 3(I)

SCOPE OF WORKS
SPECIFICATIONS

DELHI METRO RAIL CORPORATION LIMITED
NBCC Place, Bhishma Pitamah Marg
Pragati Vihar, New Delhi - 110 003 India



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DELHI METRO RAIL CORPORATION LIMITED

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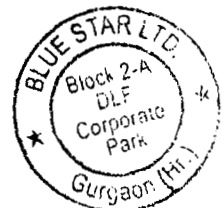
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SCOPE OF WORK - ECS

DELHI METRO RAIL CORPORATION LTD.
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MASS RAPID TRANSIT SYSTEM -PHASE-II

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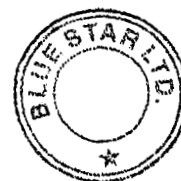
VOLUME 3

SCOPE OF WORK (PART-1)

DELHI METRO RAIL CORPORATION LIMITED
NBCC Place, Bhishma Pitamah Marg
Pragati Vihar, New Delhi - 110 003, India



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1. Introduction

1.1 Scope and Purpose

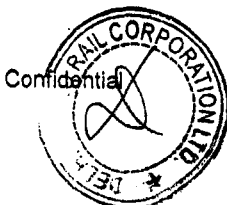
This section of specification defines the objectives, guidelines and requirements for Detailed Engineering, Manufacture, Supply, Installation, Testing and Commissioning of Environment Control System (ECS) for 7 underground stations and associated tunnel sections. The Contractor shall be responsible for the detailed design verification of station environmental control system together with layout, supply and installation and commissioning of all equipment, except tunnel ventilation, including pipe work, valves, brackets, fittings, sleeves and insulation to complete the installation.

The works to be executed under the Contract include manufacture, verification of design, delivery to site, installation, testing, including integrated testing and commissioning, technical support and documentation for a complete integrated Environment Control System (ECS) including all control centres, AHUs, ECS fans, Dampers, Ducts, Electric Panels, switchgears necessary to deliver the requirements of this Specification. It shall also include all electrical panels required for operation of the Tunnel Ventilation System (TVS) as per specification, SLD and BOQ provided including termination of cables both side.

The Environment Control System is to be manufactured, supplied, installed, tested and commissioned by the Contractor and shall meet all performance and functional requirements as defined in the Specification. The Commissioning of the full Environment Control System including all equipments e.g. Chillers, Pumps, Cooling Towers, AHUs, Fans, Electrical Panels and associated cables etc. The ECS contractor have to interface with the TVS and BMS contractor. Any failure of the Environment Control System equipment will be sole responsibility of the ECS contractor. The contractor shall interface with the TVS contractor for manufacture of TVS electrical panels, erection, testing and commissioning of the Electrical system required for the TVS.

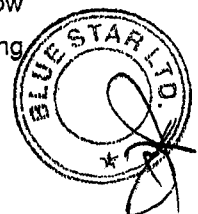
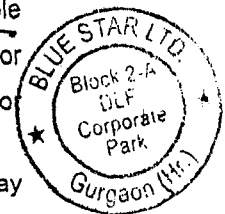
The Contractor shall verify the equipment capacities and sizing and shall be responsible to obtain the performance requirements mentioned in this scope of work. The Contractor shall also confirm the general adequacy of the space requirements within voids of service ducts, openings, main routes etc.

Trackway exhaust systems will be installed by the TVS contractor to enhance trackway temperature control by removing a portion of train generated heat while the train is within the bounds of a station. This system involves the use of fans to induce airflow across heat sources located beneath the transit vehicle as well as from air conditioning



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units on top of the transit vehicle and exhaust the heated air. During normal conditions, trackway exhaust fans shall operate continuously. When the station air conditioning system is operating in closed environment, the air collected from the trackway exhaust system shall be the return air feed to the station air handling units.

Station air conditioning systems shall maintain specific temperature and humidity conditions in underground station concourse and platform public areas. During an emergency fire condition within a station, the station air handling system shall be operable as a smoke removal system to supplement the smoke extraction system.

The contractor should adopt the internationally proven technology to ensure reliable and economical service. All equipment selected should occupy minimum space, energy efficient and having high power to weight ratio as per International Standards.

The air handling units and associated equipments smoke extraction and ventilation fans with related equipments shall be installed in the ECS plant rooms. The actual equipment positions, sizes and arrangement shall be verified by the Contractor to suit each station. Actual equipment capacities and selections shall be determined on an individual basis for each station and basis of selection to be brought out in details.

Tunnel Ventilation and ECS systems are inter-related. This aspect must be taken into account while designing the ECS system.

The station air conditioning systems shall also be operable as a smoke control system during emergency conditions. The smoke control system shall be capable of providing a stream of fresh air in station entrances; all exit routes and concourse area as well.

Performance requirements

Station air conditioning systems shall be designed to maintain specific design temperatures and provide a minimum of 0.005 m³/s fresh air per person. 10% of the total circulation air shall be added from the outside fresh air at the AHU inlet to pressurise the station.

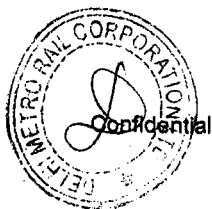
Station air conditioning system design conditions shall be:

Platform, Concourse- 27°C at 55% relative humidity

1.2 Work Included In the Services

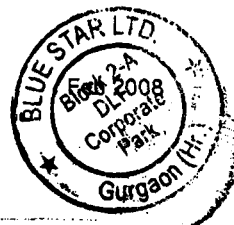
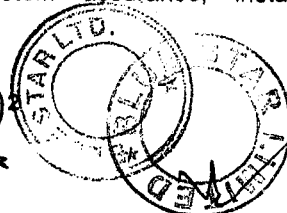
The Services to be performed by the Contractor shall include, but not be limited to, the following:

1. Manufacture, delivery, system assurance, installation, testing and commissioning of the ECS.



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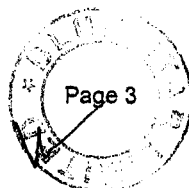
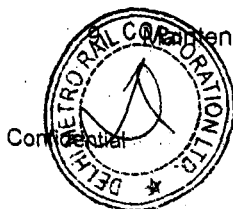
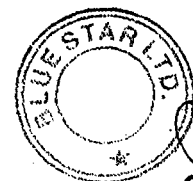
2. Manufacture, delivery, system assurance, installation, testing and commissioning of Electrical system of the TVS and ECS.
3. Technical supports for operation and maintenance of the ECS.
4. Supports for training.
5. Training of the Employer's engineers and training instructors, operations staff, maintenance staff and engineering staff.
6. Decommissioning, removal and/or disposal of temporary works.
7. Manpower and all necessary tools for work execution and installation commissioning.
8. Warranty period after commissioning.

1.2.1 In a general manner, all works, facilities and services and other components as required whether or not specified necessary to deliver the requirements of this Specification to ensure a perfect and complete execution of ECS works under this scope and according to relevant code and standards and to this Specification.

1.3 Scope of the work of supply

The Contractor shall supply all equipment and facilities necessary to meet the requirements of this technical specification, including, **but not limited to:**

1. Chillers, Cooling Towers, Condenser & Chilled water pumps, Air Handling Units, Fan Coil Units, ECS fans, Split A/c, different dampers required for ECS system, Louvers, Grilles & Diffusers, piping, Duct work, cable trays, cabling and MCC's along with DBs etc. required for efficient working of ECS system.
2. All auxiliary equipment such as: conical fan inlets and outlets, sound attenuators, ducts and inspection covers
3. All cable trays, cables and cabling necessary for the ECS and TVS Works;
4. All panels for the operation of the TVS and ECS works.
5. Enclosures and supporting brackets for housing and fixing equipment;
6. Power supply and distribution panel, circuit breakers, and isolators;
7. All equipment associated with any interfaces required to ensure operation within the performance requirements;
8. All special test equipment and tools, including data configuration tools;
9. Maintenance tools;



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